Name: _____

TOPIC TEST 1

Chapter 7 Logarithmic functions

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Four short-answer questions (9 marks).
- Part C: One analysis question (11 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Find the exponential form of the logarithmic equation $\log_3\left(\frac{1}{9}\right) = -2$.

A $3^2 = \frac{1}{9}$

B $\left(\frac{1}{9}\right)^{-2} = 3$

C $2^3 = \frac{1}{9}$

D $3^{-2} = \frac{1}{9}$

E $3^2 = 9$

[1 mark]

2 $3\log(10) - \log(100)$ when simplified is equal to:

A $\log(10)$

B $3\log(90)$

C 1

D $\log(30)$

E none of these

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

3 If $f(x) = \log_e(3x)$ then $f'(1)$ equals:

A 1

B 3

C $\frac{1}{3}$

D $\log_e(3)$

E $3\log_e(3)$

[1 mark]

4 If $\log\left(\frac{a}{b}\right) + \log\left(\frac{b}{a}\right) = \log(a + b)$ then

A $a + b = 1$

B $a = b$

C $a - b = 1$

D $a^2 + b^2 = 1$

E $a^2 - b^2 = 1$

[1 mark]

5 If $\log 27 = 1.431$, then the value of $\log 9$ is:

A 0.934

B 0.477

C 0.945

D 0.954

E 0.958

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

Part B

Short-answer questions

Four short-answer questions
9 marks
Show your working where appropriate.

1 Find $\frac{dy}{dx}$ given $y = \log_e(3x - 1)$.

[1 mark]

2 Find the coordinate on the curve $y = 6\ln(x)$ at which the gradient is 2.

[3 marks]

3 Anti-differentiate $\frac{x}{5x^2 - 2}$.

[2 marks]

4 The magnitude, M, of an earthquake is measured using the Richter scale $M = \frac{2}{3}\log_{10}\left(\frac{E}{10^{4.4}}\right)$ where M is the magnitude and E is the seismic energy released by the earthquake (in joules). On 25 September 2003, an earthquake measuring 7.6 on the Richter scale shook Hokkaido, Japan. How much energy (joules) did the earthquake release?

[3 marks]

Part C

Analysis question

One analysis question
11 marks
Show your working where appropriate.

1 A farmer has one cow with a contagious disease in a herd of 1000. If the cow is left untreated, the time in t days for n of the cows to become infected is modelled by: $t = -5\log_e\left(\frac{1000 - n}{999n}\right)$

a Find the number of days (to 1 decimal place) that it takes for the disease to spread to:

i 100 cows

ii 200 cows

iii 998 cows

iv 999 cows.

b Find how many cows will be infected after 35 days.

c Sketch this function for $0 < n \leq 1000$.

d Using this model, describe the rate at which this contagious disease spreads over time.

[11 marks]

[Total marks: 25]

© Cengage Learning Australia Pty Ltd 2014

Answers

Part A – Multiple-choice questions [1 mark each]

1 D

2 C

3 A

4 A

5 D [5 marks]

Part B – Short-answer questions

1 $\frac{dy}{dx} = \frac{3}{3x - 1}$ [1 mark]

2 $\frac{dy}{dx} = \frac{6}{x}$ [1 mark]

$2 = \frac{6}{x}$

$2x = 6$

$x = 3$ [1 mark]

$x = 3, y = 6 \ln 3$

Therefore the coordinate is $(3, 6 \ln 3)$ [1 mark]

3 Since $\int \frac{10x}{5x^2 - 2} dx = \log_e(5x^2 - 2) + c$ [1 mark]

then $\int \frac{x}{5x^2 - 2} dx = \frac{1}{10} \int \frac{10x}{5x^2 - 2} dx = \frac{1}{10} \log_e(5x^2 - 2) + c$ [1 mark]

4 $M = \frac{2}{3} \log_{10}\left(\frac{E}{10^{4.4}}\right)$

[1 mark]

$7.6 = \frac{2}{3} \log_{10}\left(\frac{E}{10^{4.4}}\right)$

$\frac{7.6 \times 3}{2} = \log_{10}\left(\frac{E}{10^{4.4}}\right)$ [1 mark]

$10^{\left(\frac{7.6 \times 3}{2}\right)} = \frac{E}{10^{4.4}}$

$10^{\left(\frac{7.6 \times 3}{2}\right)} \times 10^{4.4} = E$

$E = 6.3 \times 10^{15}$ joules [1 mark]

Part C – Analysis question

1 a i $n = 100, t = -5\log_e\left(\frac{1000 - 100}{999 \times 100}\right)$

$t = 23.5$ days

ii $n = 200, t = -5\log_e\left(\frac{1000 - 200}{999 \times 200}\right)$

$t = 27.6$ days

iii $n = 998, t = -5\log_e\left(\frac{1000 - 998}{999 \times 998}\right)$

$t = 65.6$ days

iv $n = 999, t = -5\log_e\left(\frac{1000 - 999}{999 \times 999}\right)$

$t = 69.1$ days **[4 marks]**

b $t = 35, 35 = -5\log_e\left(\frac{1000 - n}{999n}\right)$ **[1 mark]**

$$\frac{35}{-5} = \log_e\left(\frac{1000 - n}{999n}\right)$$

$$-7 = \log_e\left(\frac{1000 - n}{999n}\right)$$

$$e^{-7} = \frac{1000 - n}{999n}$$

$e^{-7} \times 999n = 1000 - n$

$999e^{-7}n + n = 1000$

$n(999e^{-7} + 1) = 1000$ **[1 mark]**

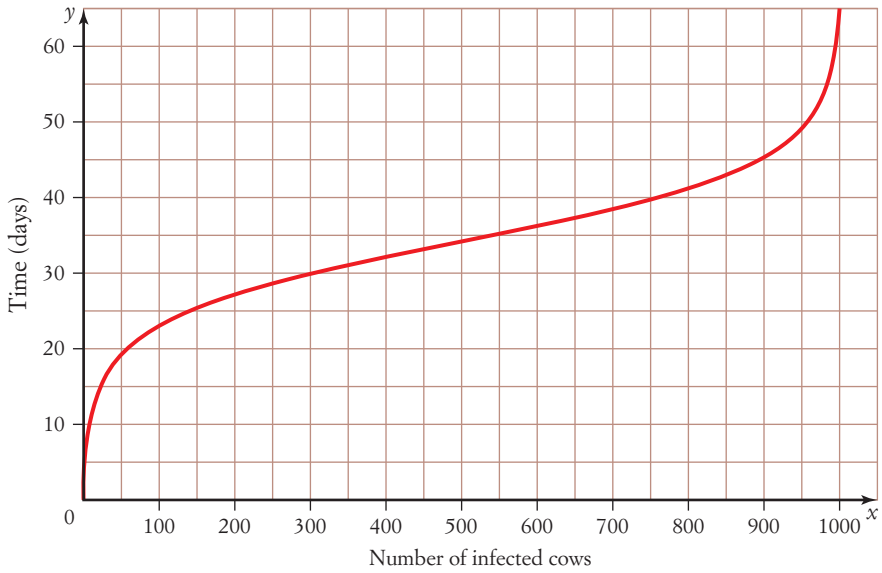
$n = \frac{1000}{999e^{-7} + 1}$ **[1 mark]**

$n = 523.3$ **[1 mark]**

After 35 days there are 523 cows infected. The 0.3 means that cow number 524 has also contracted the disease partially. Therefore, there will be 524 cows infected after 35 days.

© Cengage Learning Australia Pty Ltd 2014

c



[2 marks]

d This contagious disease can spread very rapidly at first and then very slowly as nearly all of the population has become infected. It is called a *logistic growth model*.

[1 mark]

© Cengage Learning Australia Pty Ltd 2014